
3AL Tutorial 6 2024

1. Let G be a group acting on a set X and let $a, b \in G$ and $x, y \in X$. Prove the following:

- (a) If $a * x = a * y$, then $x = y$,
- (b) $a * x = b * y$ if and only if $(b^{-1}a) * x = y$.

2. Prove that if H is a normal subgroup of G , then G acts on both H and G/H by conjugation.

NOTE: For $aH \in G/H$ and $g \in G$, we define $g(aH)g^{-1} = \{gahg^{-1} : h \in H\}$. To prove that conjugation is well-defined on G/H , you will have to show that $g(aH)g^{-1}$ is indeed an element of G/H .

3. Prove that $*$: $\text{GL}_n(\mathbb{R}) \times \mathbb{R} \rightarrow \mathbb{R}$ defined by $A * x = \det(A)x$ is a group action.
4. Construct a group action $*$: $G \times \text{Aut}(G) \rightarrow \text{Aut}(G)$, where $\text{Aut}(G)$ is the group of automorphisms of G .
5. Let G be a group acting on a set X .

- (a) Prove that, for all $x \in X$, the stabiliser $S(x)$ is a subgroup of G .
- (b) Prove that if, for $x, y \in X$, there is some $g \in G$ such that $y = g * x$, then $S(y) = gS(x)g^{-1}$.

6. Consider the additive group $\mathbb{R}$.

- (a) Prove that the map $*$: $\mathbb{R} \times \mathbb{C} \rightarrow \mathbb{C}$ defined by $a * z = ze^{ia}$, for $a \in \mathbb{R}$ and $z \in \mathbb{C}$, is a group action.
- (b) Find all stabilisers and orbits.
- (c) Geometrically, what does this action represent?

Extra problems

For a nonempty set X , the set S_X , that consists of all bijections $\sigma : X \rightarrow X$, is a group under composition.

- (I) Let G be a group acting on a set X . For each $g \in G$, we can define a map $\sigma_g : X \rightarrow X$ by $\sigma_g(x) = g * x$. Prove that σ_g is a bijection. That is, $\sigma_g \in S_X$.
- (II) Let G be a group acting on a set X . Prove that the map $\rho : G \rightarrow S_X$ defined by $\rho(g) = \sigma_g$ is a homomorphism.
- (III) Prove the following theorem:

Cayley's Theorem. *Every group G is isomorphic to a subgroup of S_G .*

HINT: Construct a group action and then use (II).